

# (OEL) – ADBMS

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**Course Title:** Advanced Database Management Systems (ADBMS)

**Assessment Type:** Open-Ended Lab (OEL)

**Total Marks:** 60 (Implementation: 10, Report: 10, Viva: 10)

## Scenario

A growing e-commerce and analytics company wants to upgrade its database infrastructure to handle large-scale data processing, high availability, and efficient querying. The company is facing issues related to data inconsistency, poor query performance, lack of scalability, inefficient indexing, and weak transaction management.

You are hired as a Database Engineer to design and implement an optimized database system using Advanced DBMS concepts.

## Tasks / Questions

### Task 1: CAP Theorem Simulation

- Simulate CAP theorem trade-offs.
- Demonstrate strong consistency and eventual consistency.
- Explain behavior during network partition.

### Task 2: MongoDB Replication

- Configure a Replica Set in MongoDB.
- Demonstrate Primary and Secondary nodes.
- Show failover mechanism and high availability.

### Task 3: Indexing Techniques

- Implement B+ Tree indexing.
- Implement Hash-based indexing.
- Compare performance for range and exact match queries.

### Task 4: Query Optimization

- Write complex queries involving joins, subqueries, and aggregation.
- Optimize queries using indexing and restructuring.
- Compare performance before and after optimization.

### Task 5: Concurrency Control

- Simulate lock-based protocol.

- Simulate timestamp ordering.
- Demonstrate conflict resolution and deadlock handling.

### **Task 6: Data Warehousing & Mining**

- Design a Star Schema.
- Perform classification and association rule mining.
- Analyze patterns from sample data.

### **Report Requirements (20 Marks)**

1. Introduction to ADBMS concepts
2. Explanation of each task
3. Screenshots of implementation
4. Performance comparison graphs/tables
5. Challenges faced and solutions
6. Conclusion and future improvements

### **Rubrics**

#### **Implementation (10 Marks)**

- Correctness: 6 Marks
- Efficiency: 2 Marks
- Completeness: 2 Marks

#### **Report (10 Marks)**

- Organization: 3 Marks
- Technical Depth: 4 Marks
- Presentation: 3 Marks

#### **Viva (10 Marks)**

- Conceptual Clarity: 5 Marks
- Confidence & Explanation: 5 Marks

### **Learning Outcomes**

- Apply advanced database concepts in real-world scenarios
- Analyze trade-offs in distributed systems
- Implement indexing and optimization techniques
- Design scalable and efficient database systems